

Roles of lifting personnel

Lifting designated person

1.1. Detailed roles

The Lifting Designated Person :

- is responsible for the implementation of the lifting standards within his/her perimeter
- identifies at the start of the project, with the project Director/Manager's approval, the resources needed to set up the project's lifting organization
- prepares or makes prepared and verify the project *Lifting Management Plan*
- provides support to the project teams for the preparation of the *Lifting Management Plan*, the *Lifting Equipment Utilization Plans* and the *Lifting Operation Plans*, when necessary
- sets up an organization for the initial examination and the periodic inspection of lifting equipment and lifting accessories
- checks the compatibility of a lifting machinery's siting/positioning on a site installation/layout plan, against the capacities of the equipment, the types of load, their volume and weight, and the site environment
- checks adequacy of measures taken for ensuring the stability of a lifting equipment in relation to the ground bearing capacity and bearing pressure
- defines the specifications for lifting operations to be endorsed by lifting subcontractor
- informs personnel involved of points of vigilance or warning signs that must be respected
- warns the management of the project of risks of failure regarding planned lifting operations
- sets up a specific organization in the event of an emergency related to lifting operations
- checks and verifies the conformity in the implementation of the project *Lifting Management Plan*, the *Lifting Equipment Utilization Plans* and the *Lifting Operation Plans*

1.2. Performance criteria

- the project/site Lifting Management Plan, the Lifting Equipment Utilization Plans and the Lifting Operation Plans are prepared and approved
- accident prevention for all personnel involved in lifting operations is included in these plans
- implementation of these plans is compliant

2. Person preparing the lifting equipment utilization plan

2.1. Detailed roles

The person preparing the *Lifting Equipment Utilization Plan* :

- prepares the *Lifting Equipment Utilization Plan*, in cooperation with Methods and Plant department of the project or their representatives at site
- produces the sizing calculation notes, calculation of pressure exerted on the ground, apply rules to be complied with as defined in the technical specifications of the lifting equipment being considered
- incorporates in the documents any site and/or contractual constraints to be considered
- positions the lifting equipment, considering existing structures and utility networks
- defines the required ground bearing capacity
- carries out a preparatory visit (site survey) to identify the specific conditions associated with the environment of the project/site
- specifies in the *Lifting Equipment Utilization Plans* the specific lifting accessories (handling fork, lifting/spreader beam, asymmetric slinging, etc.) to be used according to the type of load to be lifted
- locates areas of the project/site where materials and equipment are to be delivered, estimate the weight of the loads and check the feasibility of each handling operation against the load chart of the selected machinery
- proposes an appropriate solution if he ascertains that the equipment's capacities are insufficient to execute the expected handling and prepares the Method Statement in conjunction with the lifting equipment supplier, the Project Manager/Project Director and the Lifting Supervisor
- approves the *Lifting Operation Plans* prepared by the Lifting Supervisor or the lifting subcontractor for "specific" and "complex" lifts

2.2. Performance criteria

- execution procedures and operational instructions are applicable and applied
- the physical integrity of persons working inside area occupied by lifting equipment is ensured
- co-activity and overlapping operating zone situations are effectively managed
- conditions for use of lifting equipment are optimal

3. Lifting supervisor

3.1. Detailed roles

The Lifting Supervisor :

- is responsible for the preparation, execution and control of the lifting operations conducted by the teams he/she manages, according to the *Lifting Equipment Utilization Plans* and the *Lifting Operations Plans*
- participates to the preparation of the *Lifting Equipment Utilization Plans*, approves them and ensures they are always up to date
- supervises directly "specific" and "complex" lifts
- ensures lifting operations are properly planned and communicated
- ensures, as work progresses, that *Lifting Equipment Utilization Plans* and *Lifting Operations Plans* remain applicable and adequate between the lifting equipment, the characteristics of the load, accessories available for use and the work environment
- verifies that equipment and accessories arriving on his site correspond to the order, are in perfect working order, and that inspections (technical and periodic) have been carried out and pending issues, if any, have been resolved
- ensures all certificates and thorough examination reports are valid for all lifting equipment and maintain them on site
- ensures that all personnel involved in lifting operations are properly trained, experienced, qualified and authorised following the project requirements
- provides specific on the job training to the various personnel involved (Operators, Signallers, Slingsers),
- appoints and coordinates Signallers
- conducts pre-lift briefings to Operators, Signallers and Slingsers about the provisions set out in the *Lifting Operations Plans* with each lift
- organises the required preliminary visits before undertaking any lifting operations to consider known risks associated with the work environment (weather, space, buried and overhead utility networks...)
- ensures the setting up of a communication protocol dedicated to lifting operations and check that means made available are working properly
- prepares a Lifting Permit for "complex" lifts
- has the authority and obligation to exercise his duties and stop any lifting operation he/she considers unsafe

3.2. Performance criteria

- the maneuvers are within the capabilities of the equipment and expectations of the project/site
- *Lifting Equipment Utilization Plans* and *Lifting Operations Plans* are complied with
- physical integrity of persons impacted (directly and indirectly) by lifting operations is ensured
- safety rules are complied with
- discrepancies are reported to be processed

4. Signallers

4.1. Detailed roles

The Signaller :

- must be present for all lifting, slewing and lowering of loads. He is attired in such a way (preferably a specific high-visibility vest) that he can be distinctively identified by lifting operators
- guides the load (without any interaction – direct or indirect - with the load)
- conducts, controls and coordinates lifting operations according to the *Lifting Operation Plans*
- assigns and supervises the Slingers involved

Before authorising any lifting, the Signaller ensures that:

- conditions set out in the *Lifting Equipment Utilization Plans* and *Lifting Operations Plans* are implemented and that the type of lift has been properly assessed
- documents relating to regulatory inspections of lifting equipment and lifting accessories are compliant
- suitability examination is carried out in the presence of the Operator and co-signed by him. This examination covers :
 - the characteristics of the load
 - the lifting equipment and accessories
- the pre-use inspection of the lifting equipment and accessories has been properly undertaken
- lifting accessories have been visually inspected to ensure they are in good condition
- loads contained in baskets will not fall out during the lift
- the load is slung correctly
- barricading is in place
- the weather conditions are compatible with the provisions set out in the Lifting plan and instruction
- the environment of the maneuver is sufficiently clear that there is no risk of destabilising nearby equipment or striking other cranes or fixed objects
- ground bearing capacity are verified against pressure exerted by the lifting equipment on each of its support points (e.g. mobile crane)
- the path of the Slingers is free of obstacles during the entire lifting operation
- a briefing is conducted with the involved personnel

The Signaller does not start the lifting operation in the event of deviation from the established *Lifting Operation Plan*. He immediately notifies the Lifting Supervisor.

During the lift, the Signaller :

- always remains in communication with the Lifting Operator, relaying instructions to him, especially in the case where the Operator does not have direct line of sight of a section of the load's travel path or when executing a blind lift
- ensures that no one is under the load when raising and landing a load, as well as between the load and a fixed object or passing mobile object
- ensures that barricading is in place and complied with and always adapted to the situation

- ensures that no one touches an approaching load unless there is a gripping point intended for such purpose (handle, tag line, push-pull pole or equivalent)
- stops a lifting operation when the actual situation does not correspond to the *Lifting Operation Plan*. He notifies immediately the Lifting Supervisor

At the end of the lift, the Signaller ensures that the load is stable before giving authorization for the slings to be slackened and for the load to be detached.

The Signaller is responsible for ensuring his crews comply with the following prohibitions :

- no simultaneous lifting of multiple items not tied or bound together
- no side pulling
- no lifting of loads that are not free of the ground or its support
- no safety devices or systems are disable or bypassed/overridden

4.2. Performance criteria

- directions and orders given by standard hand signals or by radio are complied with
- the maneuvers are within the capabilities of the equipment and expectations of the project/site
- the method of execution and the *Lifting Operation Plans* are complied with
- physical integrity of persons inside the lifting exclusion zone is ensured
- safety rules are complied with
- unsafe operations are stopped or not started and the signaller communicates to the lifting supervisor the incidents to improve the lifting process.

5. Lifting operator

5.1. Detailed roles

The Lifting Operator :

- operates the lifting equipment, such as a tower/mobile crane, gantry crane, etc. and is responsible for operating the lifting equipment within its operating limits and in accordance with all guidance and rules
- operates and maintain the lifting equipment according to the manufacturer's requirements and instructions
- operates the lifting equipment according to the *Lifting Equipment Utilization Plans* and the *Lifting Operation Plans* and follows the instructions given by the Signaller
- does not start and stops any lifting operation if considered hazardous for himself/herself or for others
- does not lift loads over personnel
- follows and responds to instructions from the Signaller
- always obeys a stop signal
- always keeps load movement under control
- inspects the lifting equipment/safety device prior to use in accordance with the manufacturer's instructions and site rules
- notifies the Lifting Supervisor if adjustments or repairs are required
- follows instructions for use provided by the manufacturer of the lifting equipment
- does not bypass or override the alarms or safety system of the lifting equipment
- does not start (or stops) the lifting operation in case of deviation to the *Lifting Operation Plan*, in case of faulty lifting equipment and/or in case of immediate hazard

5.2. Performance criteria

- directions and orders given by standard hand signals or by radio are complied with
- the load is raised vertically
- execute hoist, lower, slew and travel movements simultaneously
- load sway is controlled
- the maneuvers are precise, regular and smooth
- the maneuvers are within the capabilities of the equipment and expectations of the project/site
- the method of execution and *Lifting Operation Plan* are complied with,
- physical integrity of personnel within the lifting exclusion zone is ensured
- safety rules are complied with
- lifting operation are not started or are stopped if all safety conditions are not met.

6. Slinger

6.1. Detailed roles

The Slinger executes the operation of attaching the lifting accessories to the lifting equipment and the attaching/detaching of the accessories to the load. The lifting accessories generally consist of a wire rope, chain or web sling, but may also include rigid elements such as lifting/spreader beams, lifting keys, clamps, etc.

The Slinger :

- ensures area around the load to be lifted and the load landing area are clear
- obtains the weight of the load to be lifted or evaluate it if it is not communicated/known
- assesses and determines the centre of gravity of the load for a safe stable slinging
- applies binding to loads of loose material
- selects the appropriate accessories (type, WLL) and use them (method of slinging) in accordance with the manufacturer's instructions for use, the *Lifting Operation Plan* and the *Standard Slinging* (when applicable)
- checks that lifting accessories are in good condition before use by carrying out a visual inspection to ensure there are no anomalies that may adversely affect the safety of the lifting operation (defective safety latches, significant deformations, tearing, damaged seams, broken strands, etc.)
- ensures that the appropriate means are in place to control and guide the load during lifting (tag lines, push-pull poles, gripping handle, etc.)
- controls the rotation of the load by maintaining use of tag lines and/or push sticks
- detaches the load only when given instruction by signaller, after the load is stabilized
- removes slings using appropriate safe means of access (scaffolding, MEWP, individual movable platforms) when this must be done at height
- pays attention to his own safety (struck by the load and/or the crane, getting caught between the load and the sling or the load and an object, falling load)

6.2. Performance criteria

- directions and orders given by standard hand signals or by radio are complied with
- the method of execution and *Lifting Operation Plan* are complied with
- physical integrity of persons inside the lifting exclusion zone is ensured
- safety rules are complied with
- alerts are raised immediately if any defective lifting equipment/device is found